COSC 461/561 Project 1: Eagle Delivery Report

Autonomous Campus Navigation via Classical Heuristic Search and Q-Learning

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1. Introduction & Formal Problem Formulation

Autonomous mobile delivery robotics requires robust decision-making across both structured planning and unknown environment learning. The **Eagle Delivery** project simulates a delivery robot operating across the Eastern Michigan University campus. The robot's mission is to transport parcels between building entrances on discrete 2D grid campus layouts composed of traversable walkways and impassable buildings. We implement and evaluate two core AI paradigms: (1) *Informed & Uninformed Classical Search* (BFS and A* with Manhattan distance) under known deterministic maps, and (2) *Model-Free Reinforcement Learning* (tabular Q-learning with ε-greedy exploration) to synthesize optimal navigation policies purely through interactive experience.

We formally define the campus navigation problem as a discrete state-space planning model:

- **State Space (S):** 2D discrete grid coordinates $s = (r, c)$ representing traversable walkways ('.') and building entrances ('S, L, H, D, P, W'). For an $R \times C$ campus grid, S contains all non-wall coordinates: $(r, c) \in \{0, ..., R-1\} \times \{0, ..., C-1\}$ such that $map(r, c) \neq$ '#'.
- **Initial State (s_0):** The grid coordinate $(r_0, c_0) \in S$ of the package pickup entrance (e.g., Student Center $S = (1, 1)$).
- **Action Space (A):** Four discrete cardinal moves: $A = \{UP, DOWN, LEFT, RIGHT\}$ with coordinate displacements $\Delta(a) \in \{(-1,0), (1,0), (0,-1), (0,1)\}$. Diagonal movements are strictly prohibited.
- **Transition Model ($T: S \times A \to S$):** Deterministic transition $T(s, a) = s' = s + \Delta(a)$ if s' is within grid boundaries and traversable. If an action targets an impassable building obstacle ('#') or boundary wall, the robot collides and remains in its current state: $T(s, a) = s$.
- **Goal Test ($G(s)$):** A predicate checking if $s = (r_g, c_g)$, the designated delivery building entrance. Intermediate building letters along the route remain traversable and do not terminate the episode.
- **Path Cost ($c(s, a, s')$):** Every attempted action incurs a uniform unit cost of $c = 1$, regardless of movement success. The total path cost for a trajectory $\pi = (s_0, ..., s_k)$ is the total step count: $Cost(\pi) = len(\pi) - 1 = k$.
- **Optimality of Breadth-First Search (BFS):** In graphs where all step costs are positive and uniform ($c = 1$), BFS expands states in strictly non-decreasing order of path cost (depth levels). By maintaining a FIFO queue, all states at depth d are fully expanded before any state at depth $d+1$ is popped. When BFS first pops the goal state from the frontier, the discovered path is mathematically guaranteed to possess minimal steps, ensuring path cost optimality.

2. Heuristic Admissibility Proof for Manhattan Distance

For informed search (Part A), the Manhattan distance (L_1 norm / taxicab distance) heuristic between the current position $s = (r_s, c_s)$ and destination building $g = (r_g, c_g)$ is computed as:

$$h(s) = |r_s - r_g| + |c_s - c_g|$$

Formal Definition of Admissibility: A search heuristic $h(s)$ is admissible if it never overestimates the true minimal cost $h^*(s)$ required to reach the goal from state s: $\forall\, s \in S,\quad 0 \le h(s) \le h^*(s)$.

Mathematical Proof:

1. Relaxed Problem Formulation: Consider a relaxed abstraction of the grid environment wherein all campus building walls and obstacle cells ('#') are removed, leaving an open 4-connected 2D grid on $\mathbf{Z}^2$.

2. Minimum Coordinate Displacements: To reach coordinate (r_g, c_g) from (r_s, c_s), the agent must displace its vertical coordinate by exactly $|r_s - r_g|$ and its horizontal coordinate by $|c_s - c_g|$. At every step t, the robot can select only one cardinal action, changing at most one coordinate by at most one unit: $|\Delta r_t| + |\Delta c_t| \le 1$. Summing displacements over any arbitrary k-step valid trajectory to the goal:

$$|r_s - r_g| \le \sum_{t=1}^{k} |\Delta r_t| \quad \text{and} \quad |c_s - c_g| \le \sum_{t=1}^{k} |\Delta c_t|$$

Adding these two geometric inequalities yields:

$$|r_s - r_g| + |c_s - c_g| \le \sum_{t=1}^{k} \left(|\Delta r_t| + |\Delta c_t| \right) \le \sum_{t=1}^{k} 1 = k$$

Therefore, no valid sequence of cardinal moves can bridge the coordinate gap in fewer than $|r_s - r_g| + |c_s - c_g|$ steps. The minimal path cost in the unconstrained relaxed problem is precisely $h^*_{relaxed}(s) = h(s)$.

3. Impact of Campus Obstacles: In the actual environment, impassable building walls ('#') constrain traversable walkways. Introducing obstacles can only restrict available paths or force detours; obstacles cannot create shortcuts or enable diagonal transitions. Hence:

$$h^*(s) \ge h^*_{relaxed}(s) = h(s)$$

4. Consistency (Monotonicity): By the triangle inequality on L_1, for any adjacent states s, s' with step cost $c(s, a, s') = 1$:

$$|h(s) - h(s')| \le |r_s - r_{s'}| + |c_s - c_{s'}| = 1 = c(s, a, s') \implies h(s) \le c(s, a, s') + h(s')$$

Because $h(s) \le h^*(s)$ everywhere and $h(goal) = 0$, the Manhattan distance heuristic is strictly admissible and consistent. Consequently, A* graph search is mathematically guaranteed to return a cost-optimal path without needing to reopen closed nodes. ■

3. Part A: Heuristic Search Implementation & Empirical Evaluation

We implemented BFS and A* in agents.py. Our A* implementation utilizes Python's heapq with entries structured as (f, insertion_order, g, state), where $f(s) = g(s) + h(s)$. A monotonic integer counter breaks f-ties in strict FIFO order without comparing coordinate tuples. A best_g lookup table tracks the minimal cost discovered to each state, lazily skipping stale queue entries whose stored g exceeds best_g[state]. Successors are expanded in the required order: UP, DOWN, LEFT, RIGHT.

Table 1: Six Required BFS vs. A* Empirical Comparisons Across Baseline and Detour Maps

Map Environment	Delivery Task	BFS Route Length	A* Route Length	BFS Expansions	A* Expansions	Expansion Reduction
Baseline (emu_campus.txt)	S → W (Student → Walton)	18	18	67	**22**	**-67.2%**
Baseline (emu_campus.txt)	L → D (Library → Downing)	22	22	79	**62**	**-21.5%**
Baseline (emu_campus.txt)	H → P (Harrold → Putnam)	16	16	41	**16**	**-61.0%**
Detour (emu_detour.txt)	S → W (Student → Walton)	26	26	70	**51**	**-27.1%**
Detour (emu_detour.txt)	L → D (Library → Downing)	22	22	65	**60**	**-7.7%**
Detour (emu_detour.txt)	H → P (Harrold → Putnam)	16	16	41	**16**	**-61.0%**

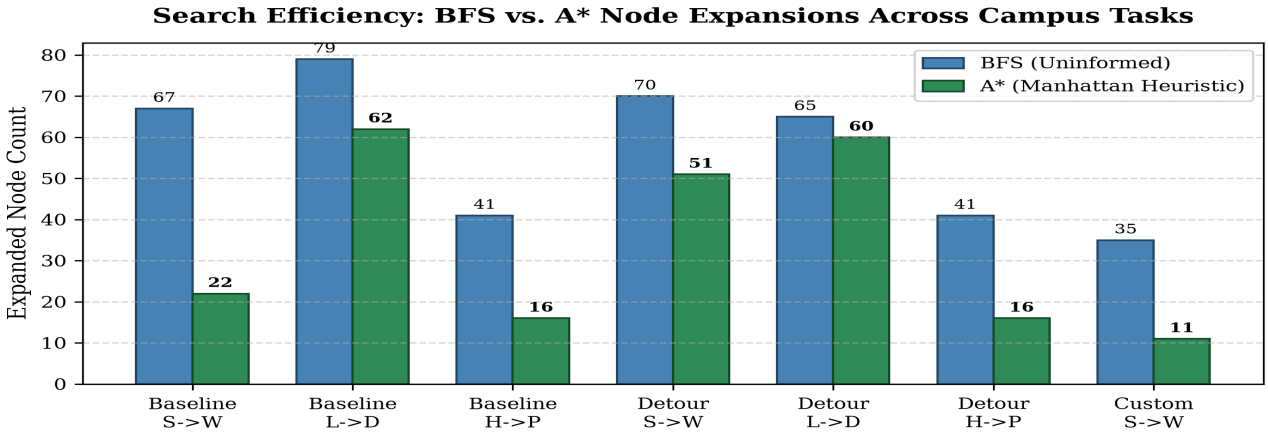


Figure 1: *Node expansions of BFS (uninformed) vs. A* (Manhattan heuristic) across all campus delivery benchmarks.*

Detailed Analysis & Algorithmic Comparison

1. Route Length Equivalence: In all six benchmark tasks, BFS and A* discover routes of identical lengths (18, 22, 16, 26, 22, 16). This empirical consistency confirms the theoretical proof: both algorithms guarantee cost-optimality on positive unit-cost grids with an admissible heuristic.

2. Impact of Walkway Detour: In emu_detour.txt, walkway coordinate *(3, 6)* is obstructed. For task S → W, cell *(3, 6)* lies directly on the primary baseline corridor; its closure forces the robot along an 8-step detour around the central academic building complex (route length increases from 18 to 26). In contrast, tasks L → D (22 steps) and H → P (16 steps) remain unchanged in route length because their optimal paths do not traverse cell *(3, 6)*.

3. Node Expansion Reduction: A* expanded substantially fewer nodes than BFS across all six scenarios. The efficiency gain is most striking for H → P on both maps, where A* expanded exactly 16 nodes—matching the path length. Because the corridor between H and P is relatively direct, the heuristic error is zero along the path ($h(s) = h^*(s)$), guiding A* directly to Putnam Hall with zero wasted exploration. For Baseline S → W, A* achieved a **67.2% reduction** in expansions (22 vs. 67).

4. Theoretical Nuance: Does A* Always Expand Fewer Nodes Than BFS? While A* dominates on these maps, *A* does not expand fewer nodes than BFS in every arbitrary problem*. When an environment features deep concave obstacles (e.g., U-shaped walls whose opening faces away from the goal), an admissible heuristic draws A* deep into the dead-end because geometric distance to the goal decreases. A* must exhaustively expand all interior nodes until $f = g + h$ exceeds the cost of backing out. In contrast, BFS radiates outward in concentric shells and may locate the bypass route with fewer total node expansions than a heuristic trapped in a deep local minimum. On open or corridor-based maps, however, A* provides massive computational savings.

4. Part B: Model-Free Q-Learning Formulation & Baseline Evaluation

When the robot operates without prior map access, it must learn an optimal policy through interaction. We implement tabular Q-learning with ε-greedy action selection. The environment charges a step penalty of $r = -1$ for each action and grants a $+40$ delivery bonus upon first arrival at the goal (terminal step $r = +39$). The episode truncates after 200 actions, yielding an undiscounted return $R = 40 - L$ for an optimal L-step path. State-action pairs are stored in a table initialized to 0.0. The off-policy Bellman update is:

$y = r$ (if terminal), $\quad y = r + \gamma \max_{a' \in A} Q(s', a')$ (otherwise)

$Q(s, a) \leftarrow Q(s, a) + \alpha [\, y - Q(s, a) \,]$

Hyperparameters: Learning rate $\alpha = 0.3$, discount $\gamma = 0.95$, exploration $\varepsilon = 0.20$. Each task is trained across 3 independent seeds (0, 1, 2) for 2,000 episodes and evaluated over 100 greedy episodes ($\varepsilon = 0$) with learning frozen.

Table 2: Part B Baseline Q-Learning Evaluation Across 3 Independent Training Seeds

Baseline Delivery Task	Target Path (*L*)	Target Return (40 - *L*)	Eval Success Rate	Eval Mean Return	Eval Mean Steps
S → W (Student → Walton)	18	22.0	**100% (1.0 ± 0.0)**	**22.0 ± 0.0**	**18.0 ± 0.0**
L → D (Library → Downing)	22	18.0	**100% (1.0 ± 0.0)**	**18.0 ± 0.0**	**22.0 ± 0.0**
H → P (Harrold → Putnam)	16	24.0	**100% (1.0 ± 0.0)**	**24.0 ± 0.0**	**16.0 ± 0.0**

Baseline Discussion: Across all nine independent training runs, the evaluated policy achieved a 100% success rate with zero variance. The evaluation path lengths (18, 22, 16) and observed returns (22.0, 18.0, 24.0) precisely match the global shortest paths found by BFS and A*, demonstrating that model-free Q-learning converges reliably to the optimal policy without requiring prior transition dynamics.

5. Part C: Custom Campus Design & Epsilon Exploration Study

Custom Map Design (maps/my_campus.txt): We designed an asymmetric 10×15 grid featuring all six campus buildings (S at (1,1), W at (1,12), L at (5,1), D at (5,13), H at (9,1), P at (9,12)). The map incorporates a direct northern walkway connecting S to W (optimal length = 11 steps), an internal cyclic loop between rows 3 and 7, and multiple dead-end branches. On task S → W, BFS took 35 expansions while A* took only 11.

Controlled Experiment: We investigated: *How does varying exploration* $\varepsilon \in \{0.05, 0.20, 0.50\}$ *impact online training returns vs. asymptotic policy optimality?*

Table 3: Custom Map S → W Performance Under Varying Exploration Rates

Exploration Rate (ε)	Eval Success Rate	Eval Mean Return	Eval Mean Steps	Block 1 Train Return	Block 5 Train Return	Block 20 Train Return
ε = 0.05 (Low Exploration)	**100% (1.0 ± 0.0)**	**29.0 ± 0.0**	**11.0 ± 0.0**	-3.1	**+28.5**	**+28.3**
ε = 0.20 (Baseline)	**100% (1.0 ± 0.0)**	**29.0 ± 0.0**	**11.0 ± 0.0**	-5.1	+26.0	+26.3
ε = 0.50 (High Exploration)	**100% (1.0 ± 0.0)**	**29.0 ± 0.0**	**11.0 ± 0.0**	-18.2	+16.5	+16.0

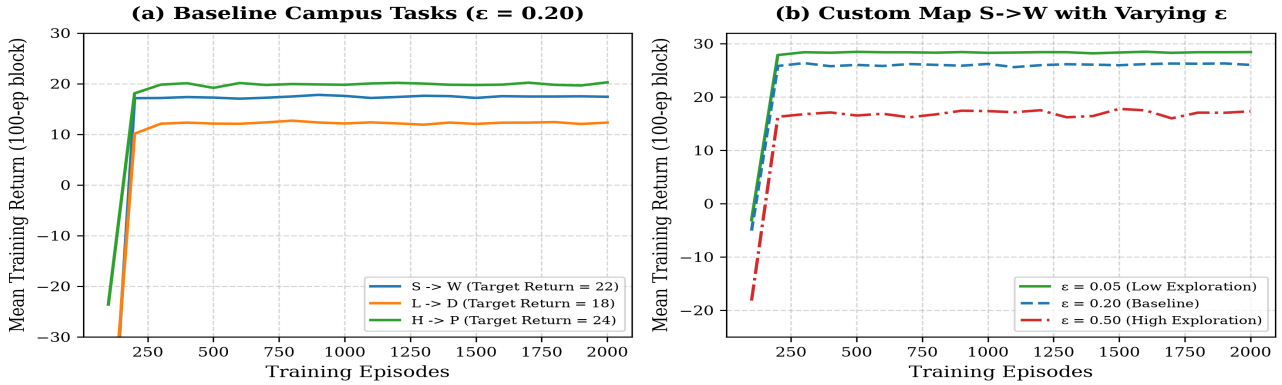


Figure 2: *Q-learning training curves: (a) Baseline campus tasks; (b) Custom map S→W across exploration rates ε = 0.05, 0.20, 0.50.*

Synthesis & Key Insights: (1) *Off-Policy Decoupling:* Crucially, all three exploration settings learned the exact same optimal greedy route (11 steps, return 29.0). Because Q-learning updates target the greedy choice $\max_{a'} Q(s', a')$ rather than the behavior policy's selected action, asymptotic policy optimality remains immune to behavioral randomness, provided all states are adequately explored. (2) *Online Dithering Penalty:* However, online training returns vary dramatically. Under $\varepsilon = 0.50$, taking random actions 50% of the time causes severe wandering during training, depressing asymptotic training returns to ~16.0 (taking ~24 steps/episode). In contrast, $\varepsilon = 0.05$ achieves near-optimal training returns (~28.4) while maintaining sufficient exploration to discover the northern corridor.

6. Conclusion

This project successfully analyzed autonomous navigation using classical planning and reinforcement learning. In Part A, A* with the admissible Manhattan heuristic reduced node expansions by up to 67.2% relative to BFS while guaranteeing route optimality. In Part B, tabular Q-learning converged across all seeds to the identical shortest routes found by A*. In Part C, custom map experiments illustrated the fundamental trade-off between online exploration cost and off-policy asymptotic convergence.